Technical report: RöstiBridge

Track 2A, UZH: Detecting Cross-Lingual Semantic Differences in Swiss Government Websites · Online · **Team RedPandas:** Christopher Yazigi, Giovanni Yazbeck, Alfredo Pio Yazbeck · **Demo:** Firefox extension `src/extension` (`dist/roestibridge-0.1.5.xpi`)

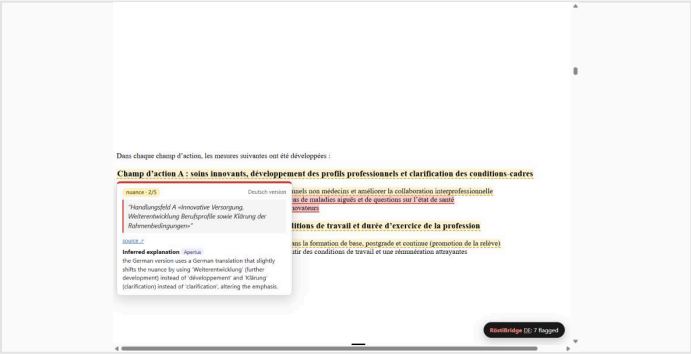
1. Summary

Swiss authorities publish the same page in up to four languages, and the versions do not always say the same thing: sentences are dropped, numbers or deadlines change, "must" becomes "may". RöstiBridge, a bridge over the *Röstigraben*, is a Firefox extension that highlights passages of an official page (federal, cantonal, RTS news or a parliamentary PDF) whose meaning may differ from the other language version. Hovering a highlight shows the other version's excerpt as a citation and an Apertus explanation in the reader's language. The same single Apertus prompt is benchmarked on SwissGov-RSD [1]: with 3-sample self-consistency, Apertus-v1.5-70B reaches a macro token-level Spearman of **0.131** on a 120-document dev subset, against 0.083 for a model-free baseline. Test predictions (168 documents) are in `predictions/test_70b_n3/`.

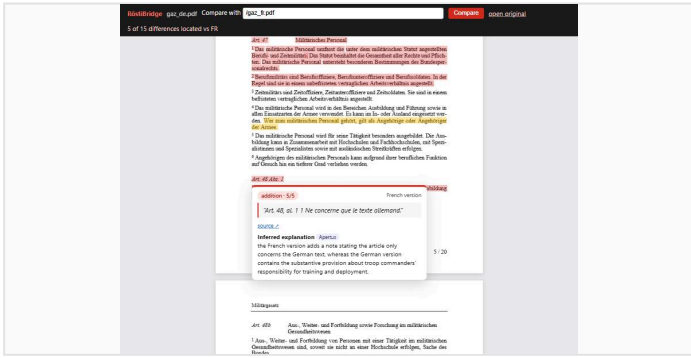
2. Architecture

```
page / PDF in language X ──► extract blocks ──► find counterpart in language Y: hreflang · switcher · /de/⇔/fr/ URL
  (content script / pdf.js)        · Apertus path matching (admin.ch CMS) · OpenParlData siblings (parlament.ch PDFs)
                                   · news RSS + Apertus story matching (RTS→SRF)
      └──► background.js: aligned chunks (≈3.5k chars, window ±4 blocks) ──► Apertus (shared/prompt.json)
           ──► JSON {a, b, type, severity, explanation} ──► filter ──► cache ──► highlight `a` + hover card (`b` + explanation)
```

- **Which version is compared** (`shared/sites.js`): federal sites with FR and DE (whichever the page is not in); bilingual cantons (BE/FR/VS/GR) with the other official language; monolingual cantons with their language.
- **Only real differences are shown:** excerpts identical on both sides, or "missing" text that is present in the other version (names, dates, numbers), are dropped. The cache is versioned.
- **Explanations** in EN / DE / FR / IT, as chosen in the options.
- **Robustness:** 120 s timeout and 3 retries per request; each section is checked on its own; only complete results are cached; the key is tested when saved.
- **Key handling:** the key stays in `browser.storage.local`, read only by the background script.
- **Benchmark pipeline** (`src/roestibridge`): stdlib LLM client, span → token projection, the official metric's pooling re-implemented, a CLI. It reads the **same** `prompt.json` (enforced by a test).



Federal page (bag.admin.ch FR vs DE): hover card



Parliamentary / Fedlex PDF (DE vs FR): hover card

3. Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-70B` (CSCS, OpenAI-compatible); `Apertus-v1.5-8B` selectable for on-prem use. **No other model is used.**
- **Uses:** (1) difference detection with severity and explanation; (2) matching translated admin.ch URL slugs; (3) RTS headline → SRF search queries and same-story decision; (4) picking the translated parliamentary PDF.
- **Prompt** `shared/prompt.json` (`rb-prompt-v1`): what counts as a difference (omission/addition, numbers/dates, obligation/scope, legal terms) and what does not (grammar, word order, free translation); exact, short excerpts; JSON output.
- **Settings:** temperature 0 (sample 1) and 0.7 (samples 2–3), `max_tokens` 2000; ≈0.82M prompt and 0.25M completion tokens for the 3-sample test run.

4. Data

- **SwissGov-RSD** [1] (`ZurichNLP/SwissGov-RSD`, CC-BY-4.0), fetched into `data/swissgov-rsd/`.
- Development on a deterministic, evenly spaced **40-document dev subset per language** (120 pairs).
- **The test split was never evaluated or inspected:** `evaluate --split test` refuses to run; test predictions were made once, with prompt and post-processing frozen on dev.
- The dataset's DE/FR/IT texts contain `�` in place of accents; the prompt asks for verbatim excerpts and the projection treats `�` as a letter.

5. Evaluation

Metric: the official one: tokens with gold ≠ −1 are pooled per pair, global Spearman, macro over EN–DE/FR/IT (matches `scipy.stats.spearmanr`).

Dev subset (40 docs/pair)	DE	FR	IT	macro
Number-mismatch baseline (no LLM)	0.102	0.068	0.080	0.083
Apertus-70B, 1 sample, raw spans	0.136	0.056	0.096	0.096
+ drop severity 1, omissions = 1.0	0.157	0.069	0.100	0.109
+ number-mismatch prior	0.159	0.074	0.104	0.112
+ 3 samples, averaged (submitted)	**0.191**	**0.077**	**0.125**	**0.131**

Ablations (`python -m roestibridge.ablate`, no API calls): binary span scores 0.095 (worse); length damping ±0.001 and paragraph smoothing 0.099 (no gain).

Error analysis (FR, 1 sample): precision is the lever (226 true vs 1,752 false positive tokens; low-severity "same meaning, different wording" spans motivated the severity ≥ 2 cut); span projection is reliable (376/382 excerpts located); FR is the hardest pair (longer documents, most ``).

Extension checks (Firefox 158 headless, WebDriver BiDi, packaged extension, live sites, 16 Oct 2026, Apertus-70B):

Case	Result
bfe.admin.ch/fr → DE	3–4 highlighted, e.g. a paragraph on the IAEA General Conference missing in German
be.ch/fr → DE (bilingual canton)	1 highlighted, a tone shift ("beau palmarès" vs "brillieren")
bag.admin.ch FR → DE	Counterpart via Apertus slug matching (`agenda-soins-de-base` → `agenda-grundversorgung`); up to 9 differences
ch.ch/it → DE and FR	Both versions compared; the homepages agree
RTS Amaudruz article → SRF	SRF story found and linked; 8–9 differences highlighted
parlament.ch PDF (DE)	Opened in the viewer; FR found via OpenParlData affair 234959; tables agree (`12'000 Franken` = `12 000 francs`)
Fedlex consultation PDF (DE↔FR)	16 differences, e.g. French "Ne concerne que le texte allemand" where German changes the provision

6. Limitations

- Document-level RSD is hard for prompted LLMs [1]; the dev subset is small (120 pairs).
- Chunk alignment is proportional (±4 blocks): very different page structures can produce "missing" reports.
- Counterpart discovery depends on site conventions; SRF is independent journalism, so RTS→SRF differences are coverage differences.
- Unsigned Manifest V2 add-on (blocking `webRequest` for PDFs).

7. Reproducibility · 8. Next steps

- `make run` (Docker, no GPU): unit tests, re-scoring of the shipped raw outputs (the numbers above, no API), baseline, rebuilt test predictions; plus a live run with `LLM_API_KEY`. `make predict-test`: 168 × 3 calls.
- Next: sentence alignment before prompting; per-language severity calibration; 8B with more samples; signed MV3 build with a "report this discrepancy to the authority" button.

License: CC-BY-4.0; pdf.js (Apache-2.0) bundled in `src/extension/pdf/lib/`. **[1]** Wastl et al. *SwissGov-RSD: A Human-annotated, Cross-lingual Benchmark for Token-level Recognition of Semantic Differences Between Related Documents.* arXiv:2512.07538 / ACL 2026.